The recurrence of OEIS A276175 takes only integer values

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Abstract

Let $a_0 = a_1 = a_2 = a_3 = 1$ and $a_n = (a_{n-1}+1)(a_{n-2}+1)(a_{n-3}+1)/a_{n-4}$ for $n \geq 4$. We prove that every a_n is an integer, settling a conjecture of B. Langlois (2016, OEIS A276175). For odd primes a short valuation induction shows that divisibility is confined to bursts of three consecutive terms. At the prime 2 we prove that $\mathrm{v}_2(a_n)$ follows an explicit word of period 44; the proof is a computer-assisted induction on eleven-step segments of the orbit, in which the deviation of each period from the initial one is confined to an explicit $\mathbb{Z}_2$-lattice tube, and the finitely many required 2-adic estimates are established by certified polynomial (jet) approximations of the segment maps. Neither of the standard integrality mechanisms applies: the Laurent phenomenon fails for this recurrence, and the map has positive algebraic entropy. The complete proof, including every certified estimate, has been formally verified.

1 Introduction

The sequence

$$1,\ 1,\ 1,\ 1,\ 8,\ 36,\ 666,\ 222111,\ 685187756,\ \ldots$$

defined by $a_0 = a_1 = a_2 = a_3 = 1$ and

$$a_n\, a_{n-4} \;=\; b_{n-1}\, b_{n-2}\, b_{n-3}, \qquad b_m := a_m + 1, \tag{1}$$

is OEIS `A276175` [1] (B. Langlois, 2016). The entry records the conjecture that all terms are integers (also asked on Math.StackExchange [2]), previously checked for $n \leq 40$. Our main result is:

Theorem 1. *$a_n \in \mathbb{Z}$ for all $n \geq 0$.*

All a_n are well defined and positive (both sides of (1) stay positive), so $a_n \in \mathbb{Q}_{>0}$. Two remarks explain why the question is not routine. First, the Laurent phenomenon fails: for generic initial values $(x_0, \ldots, x_3)$ the term a_8 has a genuine pole along $x_1 + 1 = 0$ (at $(1, -1, 1, 1)$ one gets $a_4 = 0$ and $a_8 = \infty$), so integrality is not an instance of cluster-algebra Laurentness [3] — it is special to the all-ones orbit. Second, the map is not integrable: the degrees d_n of a_n as rational functions of generic initial data satisfy $d_n = d_{n-1} + d_{n-2} + d_{n-3} - d_{n-4}$ for $n \geq 9$, so the algebraic entropy $\log \lambda$ is positive, $\lambda \approx 1.72208$ being the largest root of $x^4 - x^3 - x^2 - x + 1$; there are no conserved quantities to exploit. The recurrence belongs to the coprimeness-preserving but non-integrable class studied by Kamiya–Kanki–Mase–Tokihiro [4].

The proof splits by prime. Section 2 treats all odd primes at once by an elementary induction (Theorem 2). Section 3 reduces the remaining 2-adic statement to a periodicity assertion, the

Word Theorem (Theorem 4). Sections 4 and 5 prove the Word Theorem: an induction over 11-step segments of the orbit whose hypotheses are finitely many exact 2-adic estimates, established by certified computation. Section 6 records the formal verification.

Throughout, v_2 and v_q denote the 2-adic and q-adic valuations on $\mathbb{Q}^\times$, and

$$\mathcal{O} \;:=\; \{x \in \mathbb{Q} : \; \mathrm{v}_2(x) \geq 0\}$$

is the ring of 2-integral rationals ($\mathbb{Z}$ localized away from odd primes).

2 Odd primes: burst confinement

Theorem 2. *For every odd prime q and every $n \geq 0$, $\mathrm{v}_q(a_n) \geq 0$.*

The proof uses only the identity obtained from (1) by adding a_{n-4} to both sides:

$$b_n\, a_{n-4} \;=\; a_{n-4} + b_{n-1}b_{n-2}b_{n-3} \qquad (n \geq 4). \tag{2}$$

Fix an odd prime q and write $\mathrm{v} := \mathrm{v}_q$. Define

$$t_n := \mathrm{v}(b_n) \;\; (0 \leq n \leq 3), \qquad t_n := \mathrm{v}(b_n) - t_{n-4} \;\; (n \geq 4),$$

so $t_0 = t_1 = t_2 = t_3 = \mathrm{v}(2) = 0$.

Lemma 3. *For all $n \geq 4$:* (I) $t_n \geq 0$; (II) $\mathrm{v}(a_n) = t_{n-1} + t_{n-2} + t_{n-3}$; (III) *if* $t_n > 0$ *then* $t_{n-1} = t_{n-2} = t_{n-3} = 0$.

Proof. Strong induction on n. For $4 \leq n \leq 7$ one has $t_{n-4} = 0$, so $t_n = \mathrm{v}(b_n) \geq 0$, and (II), (III) are read off from $a_4 = 8$, $a_5 = 36$, $a_6 = 666$, $a_7 = 222111$ and $b_4 = 3^2$, $b_5 = 37$, $b_6 = 23 \cdot 29$, $b_7 = 2^5 \cdot 11 \cdot 631$. Let $n \geq 8$ and assume (I)–(III) below n.

(II). By (1) and the definition of t,

$$\mathrm{v}(a_n) = \sum_{i=1}^{3} \mathrm{v}(b_{n-i}) - \mathrm{v}(a_{n-4}) = \sum_{i=1}^{3} (t_{n-i} + t_{n-i-4}) - (t_{n-5} + t_{n-6} + t_{n-7}) = t_{n-1} + t_{n-2} + t_{n-3},$$

using (II) at $n-4$; the sum telescopes and no inequality is needed.

(I). We must show $\mathrm{v}(b_n) \geq t_{n-4}$. If $t_{n-4} = 0$ this holds because $\mathrm{v}(b_n) = \mathrm{v}(1 + a_n) \geq \min(0, \mathrm{v}(a_n)) \geq 0$ by (I), (II) below n. So assume $T := t_{n-4} > 0$. By (III) at $n-4$ we get $t_{n-5} = t_{n-6} = t_{n-7} = 0$, hence $\mathrm{v}(a_{n-4}) = 0$ by (II); and for $i = 1, 2, 3$ the sum in (II) for $\mathrm{v}(a_{n-i})$ contains the term t_{n-4}, so $\mathrm{v}(a_{n-i}) \geq T$, i.e. $b_{n-i} \equiv 1 \pmod{q^T}$. Then (2) gives

$$b_n\, a_{n-4} \;\equiv\; a_{n-4} + 1 \;=\; b_{n-4} \pmod{q^T},$$

and $\mathrm{v}(b_{n-4}) = t_{n-4} + t_{n-8} \geq T$, so $\mathrm{v}(b_n a_{n-4}) \geq T$; since $\mathrm{v}(a_{n-4}) = 0$, this yields $\mathrm{v}(b_n) \geq T$.

(III). If $t_n > 0$ then $\mathrm{v}(b_n) > 0$, so $a_n \equiv -1 \pmod{q}$ and $\mathrm{v}(a_n) = 0$; by (II) and (I), $t_{n-1} = t_{n-2} = t_{n-3} = 0$. ☐

Theorem 2 follows from Lemma 3(I)–(II). Divisibility by an odd prime is thus confined to *bursts*: a positive t_n forces the three preceding t's to vanish, and $\mathrm{v}_q(a_n)$ is always the sum of the three preceding t's. This is the valuation shadow of the confined singularity pattern of the map.

3 Reduction to the Word Theorem

By Theorem 2, Theorem 1 is equivalent to $\alpha_n := \mathrm{v}_2(a_n) \geq 0$ for all n. Let $\beta_n := \mathrm{v}_2(b_n)$; by (1),

$$\alpha_n = \beta_{n-1} + \beta_{n-2} + \beta_{n-3} - \alpha_{n-4}. \tag{3}$$

Theorem 4 (Word Theorem)**.** *For all $n \geq 4$, $\alpha_n = w\big((n-4) \bmod 44\big)$, where w is the word*

$3,2,1,0,2,3,4,0,0,0,0,\ \ 3,2,1,0,2,3,4,0,0,0,0,\ \ 4,3,2,0,1,2,3,0,0,0,0,\ \ 4,3,2,0,1,2,3,0,0,0,0.$

Every letter of w is nonnegative, so Theorem 4 implies Theorem 1. (The proof also establishes the accompanying period-44 word for β_n, namely $0,0,0,5,0,0,0,2,1,1,1,0,0,0,5,0,0,0,2,1,1,2,0,0,0,5,0,0,0,1,1,1,2,0,0,0,5,0,0,0,1,1,1,1$; consistency: $\sum w = \frac{3}{2}\sum\beta$ over a period, forced by (3). The block structure $PP\bar{P}\bar{P}$ of w reflects the palindrome symmetry $a_{3-n} = a_n$ of the doubly infinite orbit.)

The base of the induction is a finite computation:

Proposition 5. *Theorem 4 holds for $4 \leq n \leq 69$. Moreover, for these n, writing $a_n = 2^{\alpha_n} u_n$ with u_n a 2-adic unit, the residues $u_n \bmod 2^{K_n}$ are known exactly for precisions $K_n \geq 558$.*

Method. Exact rational arithmetic verifies the word for $n \leq 28$. Beyond that one computes in $\mathbb{Q}_2$ with explicit precision tracking: values are stored as $2^v u$ with u known modulo 2^π; multiplication and division cost no relative precision, while adding 1 at a β-event costs exactly β bits. Starting from 600 bits at the seed, the run to $n = 69$ retains ≥ 558 bits, far above the ≤ 6 bits needed to certify each event. (The same computation extends to $n = 30{,}000$ with 4,900 bits retained, which is how the word was found; only $n \leq 69$ is used below.) ☐

4 Proof of the Word Theorem

Windows, tubes, and the segment maps

Fix the four *boundaries* $B_i = 22 + 11i$, $i = 0,1,2,3$, so that $B_3 + 11 = B_0 + 44$. All indices $22 \leq n \leq 69$ carry certified unit residues by Proposition 5; note $\alpha_{B_i+j} = 0$ for $j = 0,\dots,3$, so the window values a_{B_i+j} are 2-adic units times 1.

For each i the certificate specifies an explicit matrix $G_i \in M_4(\mathbb{Q}) \cap M_4(\mathcal{O})$ (its *tube generators*; every entry is an odd integer times a power of 2, every nonzero entry of every column having $\mathrm{v}_2 \geq 3$), which is invertible with *dyadic* inverse: the matrix $H_i := G_i^{-1}$ has entries in $\mathbb{Z}[\frac{1}{2}]$ with $\mathrm{v}_2 \geq -5$. The *tube* at boundary i is the $\mathcal{O}$-span of the rows of G_i,

$$V_i := \Big\{\xi \in \mathcal{O}^4 :\ \xi_j = \sum_{a=0}^{3} c_a\,(G_i)_{aj} \text{ for some } c \in \mathcal{O}^4\Big\} = \Big\{\xi :\ \textstyle\sum_j (H_i)_{aj}\,\xi_j \in \mathcal{O}\ \ (a = 0,\dots,3)\Big\},$$

a sub-$\mathcal{O}$-module of $2^3\mathcal{O}^4$. (The second description follows from $H_i = G_i^{-1}$ and is how membership is read off; the two-sidedness of the inverse is a finite rational identity.)

For $c \in \mathcal{O}^4$ define the *perturbed segment orbit* at boundary i:

$$F_j^{(i,c)} := a_{B_i+j}\Big(1 + \sum_a c_a (G_i)_{aj}\Big)\ \ (j = 0,\dots,3), \qquad F_m^{(i,c)} := \frac{(F_{m-1}^{(i,c)}+1)(F_{m-2}^{(i,c)}+1)(F_{m-3}^{(i,c)}+1)}{F_{m-4}^{(i,c)}}\ \ (m \geq 4).$$

Thus $F_m^{(i,0)} = a_{B_i+m}$, and if the actual window at $B_i + 44k$ is a perturbation of the base window with deviation in V_i — that is, $a_{B_i+44k+j} = F_j^{(i,c)}$ for some $c \in \mathcal{O}^4$ — then $F_m^{(i,c)} = a_{B_i+44k+m}$ for *all* m, since both sides satisfy the same recurrence from the same four initial values.

The computational core of the proof is the following statement, one instance per boundary.

Lemma 6 (Segment Lemma)**.** *For each $i \in \{0, 1, 2, 3\}$ and every $c \in \mathcal{O}^4$:*

(i) *for $m = 4, \ldots, 14$: $F_m^{(i,c)} \neq 0$ and $\mathrm{v}_2\big(F_m^{(i,c)}\big) = w\big((B_i + m - 4) \bmod 44\big)$;*

(ii) *for $a = 0, \ldots, 3$:* $$\sum_{j=0}^{3} (H_{i+1})_{aj} \Big(\frac{F_{11+j}^{(i,c)}}{F_{11+j}^{(i,0)}} - 1 \Big) \ \in \ \mathcal{O},$$

where H_{i+1} is taken with $i+1$ modulo 4. In particular, by the dual description of the tubes, the output deviation $\big(F_{11+j}^{(i,c)}/a_{B_i+11+j} - 1\big)_j$ lies in V_{i+1}.

The proof of Lemma 6 is a finite certified computation, described in Section 5. We first show how it yields the theorem.

The induction over periods

For $i \leq 2$ we have $B_i + 11 = B_{i+1}$, so $F_{11+j}^{(i,0)} = a_{B_{i+1}+j}$ and Lemma 6(ii) transports a tube perturbation at boundary i directly to one at boundary $i+1$, with the same k: if $a_{B_i+44k+j} = F_j^{(i,c)}$ ($j \leq 3$) with $c \in \mathcal{O}^4$, then $a_{B_{i+1}+44k+j} = F_j^{(i+1,c')}$ with $c'_a = \sum_j (H_{i+1})_{aj} (a_{B_{i+1}+44k+j}/a_{B_{i+1}+j} - 1) \in \mathcal{O}$, using $G_{i+1}H_{i+1}$-duality to reconstruct the deviation from its coordinates.

At the wrap $i = 3$ the centers differ by one period: $F_{11+j}^{(3,0)} = a_{66+j} = a_{B_0+44+j}$, while the next tube is anchored at a_{22+j}. Write

$$\xi'_j := \frac{a_{66+44k+j}}{a_{66+j}} - 1, \qquad \xi^0_j := \frac{a_{66+j}}{a_{22+j}} - 1, \qquad \xi^{\mathrm{tot}}_j := \frac{a_{22+44(k+1)+j}}{a_{22+j}} - 1,$$

so that $1 + \xi^{\mathrm{tot}}_j = (1 + \xi'_j)(1 + \xi^0_j)$, i.e. $\xi^{\mathrm{tot}} = \xi' + \xi^0 + \xi'\xi^0$ componentwise. Lemma 6(ii) gives $H_0\xi' \in \mathcal{O}^4$. The vector ξ^0 is a fixed rational vector; from the certified residues of Proposition 5 one verifies
$$\mathrm{v}_2(\xi^0) = (5, 4, 4, 5), \qquad H_0\, \xi^0 \in \mathcal{O}^4 \tag{4}$$
(the four coordinates of $H_0\xi^0$ have $\mathrm{v}_2 = (1, 4, 0, 1)$; the individual products $(H_0)_{aj}\xi^0_j$ can have $\mathrm{v}_2 = -1$, so the cancellation across j is genuine and is checked on the combined values). Finally, for the cross term: $\xi' \in V_0$ gives $\mathrm{v}_2(\xi'_j) \geq 4$ (the minimal column valuation of G_0), so with $\mathrm{v}_2((H_0)_{aj}) \geq -5$ and (4) each product $(H_0)_{aj}\, \xi'_j\, \xi^0_j$ has $\mathrm{v}_2 \geq -5 + 4 + 4 = 3$. Summing, $H_0\, \xi^{\mathrm{tot}} \in \mathcal{O}^4$, i.e. $\xi^{\mathrm{tot}} \in V_0$: a tube perturbation at (B_3, k) transports to one at $(B_0, k+1)$.

Proof of Theorem 4. By induction on k: the window at $B_0 + 44k$ is a V_0-perturbation of the base window. For $k = 0$ take $c = 0$. Given the statement at k, apply Lemma 6 for $i = 0, 1, 2$ and the wrap argument for $i = 3$; along the way part (i) yields $\alpha_{B_i+44k+m} = w((B_i + m - 4) \bmod 44)$ for $m = 4, \ldots, 14$ and all four i, i.e. the word on the 44 consecutive indices $26 + 44k, \ldots, 69 + 44k$. Ranging over k this covers all $n \geq 26$; the range $4 \leq n \leq 25$ is Proposition 5. $\square$

5 Certification of the Segment Lemma

Lemma 6 quantifies over the infinite set $c \in \mathcal{O}^4$; it is established by computing with certified polynomial enclosures (*jets*) of the functions $c \mapsto F_m^{(i,c)}$. All arithmetic below is exact rational arithmetic; every bound is a 2-adic valuation inequality between explicitly given rational numbers. We write $F_m := F_m^{(i,c)}$ and suppress i.

Definition 7. A *jet* is a pair $J = (P, \tau)$ with $P \in \mathbb{Q}[x_0, x_1, x_2, x_3]$ given by its (sparse) coefficient list and $\tau \in \mathbb{Z}$. It *represents* a function $f : \mathcal{O}^4 \to \mathbb{Q}$ if

(a) $\mathrm{v}_2\big(f(c) - P(c)\big) \geq \tau$ for all $c \in \mathcal{O}^4$, and

(b) for all $c, c' \in \mathcal{O}^4$ and integers $e \geq 0$ with $\mathrm{v}_2(c'_a - c_a) \geq e$ for each a:

$$\mathrm{v}_2\Big(\big(f(c') - P(c')\big) - \big(f(c) - P(c)\big)\Big) \;\geq\; \tau + e.$$

Condition (b) — the *two-point* tail bound — is what makes truncation sound: it expresses that the neglected tail is itself a 2-adically Lipschitz function of the parameters, which is automatic for polynomial tails (every monomial difference retains a difference factor) and is preserved by all operations below.

Write $\delta(P) := \min_t \mathrm{v}_2(\text{coefficient } t)$ over the terms of P. The following closure rules for jets in the sense of Definition 7 are elementary ultrametric estimates; each is stated with the exact parameter update used.

Lemma 8. *Let (P_1, τ_1), (P_2, τ_2) represent f, g. Then:*

(1) (product) $(P_1P_2,\ \min(\tau_1 + \min(\delta(P_2), \tau_2),\ \delta(P_1) + \tau_2))$ *represents fg;*

(2) (reanchoring) *if $P_1 = P' + E$ with $\delta(E) \geq s$, then $(P', \min(\tau_1, s))$ represents f; likewise scalar multiples and sums with the obvious updates;*

(3) (certified inversion) *if f takes unit values on $\mathcal{O}^4$* ($\mathrm{v}_2(f(c)) = 0$), *$\delta(P_1) \geq 0$, $\tau_1 \geq 0$, and a* witness *polynomial W satisfies $\delta(P_1W - 1) \geq s$, then $(W,\ \min(s, \tau_1 + \delta(W)))$ represents $1/f$;*

(4) (event reading) *if $f = F/2^\alpha$ for a represented window value F with constant term $T = P_1(0)$, and the explicit checks* $\mathrm{v}_2(1+T) = \beta$, *$\delta(P_1 - T) \geq \beta + 1$ and $\tau_1 \geq \beta + 1$ hold, then $F + 1 \neq 0$ and* $\mathrm{v}_2(F(c) + 1) = \beta$ for every *$c \in \mathcal{O}^4$, and $((1 + P_1)2^{-\beta},\ \tau_1 - \beta)$ represents $(F + 1)/2^\beta$;*

(5) (division step) *valuations add along* (1)*: if the three b-readings and the divisor have certified valuations $\beta', \beta'', \beta''', \alpha'$ on all of $\mathcal{O}^4$, the new value is nonzero with valuation $\beta' + \beta'' + \beta''' - \alpha'$ on all of $\mathcal{O}^4$, and its unit part is represented by the product jet of the three unit b-jets and the inverted divisor jet;*

(6) (deviation reading) *if (P, τ) represents the unit part f of an output value, $P = T + R$ with constant T odd and $R(0) = 0$, $\delta(R) \geq 0$, then for all $c \in \mathcal{O}^4$*

$$\mathrm{v}_2\Big(T\big(\tfrac{f(c)}{f(0)} - 1\big) - R(c)\Big) \;\geq\; \tau.$$

Proof sketch. (1): $fg - P_1P_2 = (f-P_1)g + P_1(g-P_2)$, bound g by $\min(\delta(P_2), \tau_2)$ on $\mathcal{O}^4$ and P_1 by $\delta(P_1)$; the two-point form follows from the same split plus (b) for the factors. (3): $1/f - W = (1-fW)/f = \big((1-P_1W)-(f-P_1)W\big)/f$ with $\mathrm{v}_2(f) = 0$. (4): $1+P_1(c) = (1+T)+(P_1(c)-T)$ and the deviation terms all have $\mathrm{v}_2 \geq \beta+1 > \beta = \mathrm{v}_2(1+T)$; then divide by 2^β. (6): with $A(c) := f(c) - P(c)$,

$$T\Big(\frac{f(c)}{f(0)} - 1\Big) - R(c) = \frac{T\big(A(c)-A(0)\big) - R(c)\,A(0)}{f(0)},$$

whose numerator has $\mathrm{v}_2 \geq \tau$ by (b) (taken at the pair $(0,c)$, $e=0$) and (a), and whose denominator is a unit. The remaining items are similar one-line computations. □

The certificate

For each boundary i the certificate consists of explicit data whose finite side conditions imply Lemma 6 by composing the rules of Lemma 8 along the eleven steps:

- *Initial jets.* For $j = 0, \ldots, 3$ the affine polynomial $T_j + T_j \sum_a x_a (G_i)_{aj}$, with T_j the certified residue of u_{B_i+j} from Proposition 5, represents the unit part of F_j with tail $\tau = K_{B_i+j}$ (≥ 558): this is exactly the statement that $u_{B_i+j} \equiv T_j \pmod{2^K}$, and the actual value a_{B_i+j} never appears — only its residue.

- *Per step* $m = 4, \ldots, 14$: one certified inversion witness (rule (3)); three products (rule (1)); after each product a reanchoring (rule (2)) onto an explicitly listed compact polynomial — the discarded part E is an exact polynomial identity, and only the valuations of its coefficients enter; an event reading (rule (4)) or, when $\alpha > 0$, the corresponding positive case, for each of the three b-factors; and the division step (rule (5)). This proves part (i) of the Lemma at step m.

- *Truncation parameters.* Kept polynomials are capped in total degree — degree 2 suffices at 36 of the 44 steps; the remaining eight need degrees 3–5 (degree 5 at exactly two steps); kept coefficients are truncated 2-adically at 90 bits, the truncation error being absorbed into the tails, which never need to exceed ≈ 15 bits. That degree 5 is necessary (a uniform degree-4 certificate fails by one bit in the tail transport) reflects genuine high-order cancellations in the dynamics.

- *Output readings.* For the four output jets ($m = 11, \ldots, 14$), rule (6) plus the explicit coefficient checks that each combined polynomial $\sum_j (H_{i+1})_{aj} T_j^{-1} R_j$ has all coefficient valuations ≥ 0 — computed on the *combined* polynomial, since the per-j summands can fall one bit short — prove part (ii). The wrap data (4) is verified directly from the residues of Proposition 5: ξ_j^0 agrees with the residue ratio $T_{66+j}/T_{22+j} - 1$ to within 2^{-500}, so its valuation and the four combined sums $H_0\xi^0$ are determined by exact rational numbers.

All side conditions are finitely many assertions of two kinds: exact polynomial identities over $\mathbb{Q}$ (each reanchoring and each inversion witness), and 2-adic valuations of explicitly given rationals. The full certificate comprises roughly 250 polynomial identities and 3,000 coefficient valuations; it is generated deterministically by short scripts and, more importantly, has been verified formally (Section 6).

Remark 9. The structure of the proof mirrors the structure of the dynamics. In logarithmic unit coordinates the 44-step return map is a 2-adic isometry (its Jacobian, suitably rescaled, lies in $GL_4(\mathbb{Z}_2)$ and is unipotent modulo 2), so consecutive-period deviations neither grow nor decay: the sequence $\mathrm{v}_2(u_{n+44}/u_n - 1)$ is *exactly* periodic in n, with zero margin over the required event depths at two of the four $\beta = 5$ events per period. The tubes V_i encode, besides the depth profile, the finitely many linear congruences ("hidden invariants") that the plain profile ball does not see; that the certificate closes only at jet degree 5 corresponds to the fact, visible in the Mahler difference hierarchy of the orbit, that the relevant cancellations are of genuinely high order.

6 Formal verification

The complete proof of Theorem 1 — Theorem 2, the base computation of Proposition 5, the Segment Lemma with its entire certificate, and the induction of Section 4 — has been formalized and machine-checked in Lean 4 with Mathlib. The formalization is a single self-contained file, `A276175_complete.lean`, in which the sequence is defined by the recurrence alone and the final statement is

```
theorem a276175_integrality (n : ℕ) : ∃ z : ℤ, a n = z
```

It compiles in under five minutes, contains no unproven assertions (`sorry`-free), and depends only on the three standard axioms of Lean's logic (propositional extensionality, choice, quotient soundness), as reported by the kernel's axiom audit. The soundness rules of Lemma 8 are proved once as reusable lemmas; the certificate data (tube generators, residues, jets, witnesses) is emitted by short, deterministic generation scripts and every side condition is discharged by kernel-checked computation.

References

[1] OEIS Foundation Inc., entry *A276175*, `https://oeis.org/A276175`.

[2] *Is A276175 integer-only?*, Mathematics Stack Exchange, question 1905063 (2016).

[3] S. Fomin and A. Zelevinsky, *The Laurent phenomenon*, Adv. in Appl. Math. **28** (2002), 119–144.

[4] R. Kamiya, M. Kanki, T. Mase, and T. Tokihiro, *Coprimeness-preserving non-integrable extension to the two-dimensional discrete Toda lattice equation*, J. Math. Phys. **58** (2017), 012702, and related works.